

Fourier Series and Fourier Transform

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1 INTRODUCTION

The primary objective of this laboratory exercise is to mathematically analyze and computationally simulate continuous-time signals using Fourier Series and Fourier Transforms. Analyzing signals in the frequency domain shows the individual sinusoidal frequency components that make up complex waveforms [1].

The exponential Fourier Series specifically decomposes continuous-time periodic signals into an infinite sum of complex exponential functions [2]. Reconstructing a signal using only a finite number of exponential terms results in waveform that exhibits oscillations near discontinuities which is a behavior known as the Gibbs phenomenon [3].

Meanwhile for aperiodic signals the Fourier Transform extends to analyzing the signal over a continuous spectrum of frequencies instead of the discrete harmonics [1]. Applying Fourier Transform to a system's impulse response gives its transfer function. The transfer function dictates how the system filters different input frequencies [2]. Moreover, the operational boundary of this filter is defined by 3dB cut-off frequency that marks the exact point where the signal's power is reduced by half [4].

Several specific MATLAB functions are required to computationally implement and visualize these mathematical operations. The syms and assume functions establish the foundational symbolic variables and their specific mathematical constraints for calculus operations. The int() function calculates the exact analytical integral to find the Fourier coefficients, while sum() and subs() are combined to evaluate and add the truncated series terms. Finally, the fourier() command computes the continuous transfer function, and plotting tools like stem(), plot(), and fplot() construct the visual representations of the signal magnitudes and cut-off relationships.

2 EXPERIMENTAL DESIGN

This section outlines the procedural framework used to decompose periodic signals and analyze system impulse responses using MATLAB's computational tools. The design is structured to

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ensure that the mathematical transition from the time domain to the frequency domain is accurate and replicable.

2.1 Signal Modeling and Fourier Series Decomposition

The first phase of the experiment focuses on representing a periodic sawtooth signal, $x(t)$, through an Exponential Fourier Series. Based on the visual characteristics of the signal provided in the laboratory specifications, the period is identified as $T = 0.75$, which establishes the fundamental frequency as shown in Equation 1:

$$\omega_0 = \frac{2\pi}{T} \quad (1)$$

To reconstruct the signal, the exponential Fourier coefficients (C_n) must be calculated. The methodology employs the following integral expression evaluated over one period:

$$C_n = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) e^{-jn\omega_0 t} dt \quad (2)$$

The experimental setup utilizes MATLAB's Symbolic Math Toolbox to define the signal $x(t) = -2t$ and solve the integral for a harmonic range of n from -10 to 10 . A critical component of this design involves handling numerical singularities; an infinitesimal constant (eps) is added to the frequency index when $n = 0$ to prevent division-by-zero errors during numerical evaluation.

2.2 System Response and Frequency Analysis

The second phase involves characterizing a system defined by its impulse response. The system is modeled by the function $h(t) = \frac{a}{2} e^{-a|t|}$ where $a \geq 0$. The experimental design uses the fourier function in MATLAB to determine the transfer function, $H(\omega)$, in the frequency domain.

A key performance metric in this design is the 3dB cut-off frequency (ω_c), defined as the point where the magnitude of the transfer function drops to $1/\sqrt{2}$ of its maximum value. The methodology requires solving the following magnitude equation to obtain a symbolic expression for ω_c :

$$|H(\omega)| = \frac{1}{\sqrt{2}} \quad (3)$$

2.3 Step-by-Step Procedure

To ensure the replicability of the results, the following procedural steps are implemented:

- 1) **Symbolic Initialization:** Define t , n , ω , and a as symbolic variables within the MATLAB workspace.
- 2) **Signal Construction:** Define the periodic signal $x(t)$ and extend it to three full periods using `piecewise` or `heaviside` functions to observe signal behavior across the interval $-1 \leq t \leq 1$.
- 3) **Coefficient Calculation:** Utilize the `int` function to compute the symbolic expression for the Fourier coefficients C_n .
- 4) **Numerical Reconstruction:** Convert symbolic coefficients to numerical values and reconstruct the signal $x_{pN}(t)$ by summing the harmonics from $n = -10$ to 10.
- 5) **Signal Comparison:** Plot the original signal against the truncated Fourier series using a time increment of $\Delta t = 1/1000$ to visualize the approximation and the effects of the Gibbs phenomenon.
- 6) **System Characterization:** Define the impulse response $h(t)$, compute the Fourier Transform $H(\omega)$, and solve for the cut-off frequency ω_c .
- 7) **Parametric Sweep:** Generate magnitude response plots for $|H(\omega)|$ across varying system parameters $a = [10, 50, 80, 100]$ to observe bandwidth variations.

2.4 Computational Logic

The programmatic flow of the experiment is summarized in the following logic:

- Initialize symbolic environment and define fundamental signal parameters (T , ω_0).
- Evaluate the integral for C_n and store coefficients in a vector array.
- Loop through the harmonic indices to compute the partial sum of the Fourier series.
- Apply the Fourier Transform to the system impulse response and solve for ω_c using the `solve` command.
- Visualize the magnitude of C_n , the reconstructed time-domain signal, and the frequency-domain magnitude responses.

3 ANALYSIS OF RESULTS

The results obtained from this exercise provide a comprehensive understanding of frequency-domain representations and system filtering behavior. To ensure computational reliability and mathematical validity of the exercise, independent simulations are consolidated and verified.

3.1 Fourier Series Decomposition and Magnitude Spectrum

A periodic signal can be broken down into individual frequency components having the magnitude spectrum showing how the signal's energy is distributed across the mentioned frequencies. The frequency-domain representation of the given periodic sawtooth signal $x_p(t) = -2t$ with a period of $T = 0.75$ quantifies energy distribution. Through calculating the exponential Fourier series coefficients (C_n) the analytical composition of the signal is found. The resulting magnitude spectrum which was plotted for the harmonic index $-10 \leq n \leq 10$ reveals a symmetric decay centered around the fundamental DC component. This peak shows that the lowest frequencies hold the majority of the

waveform's power while the higher harmonics have less. Just to maintain computational stability throughout numerical evaluation of this an infinitesimal variable (`eps`) was required to prevent division-by-zero singularities at the exact zero-frequency index ($n = 0$).

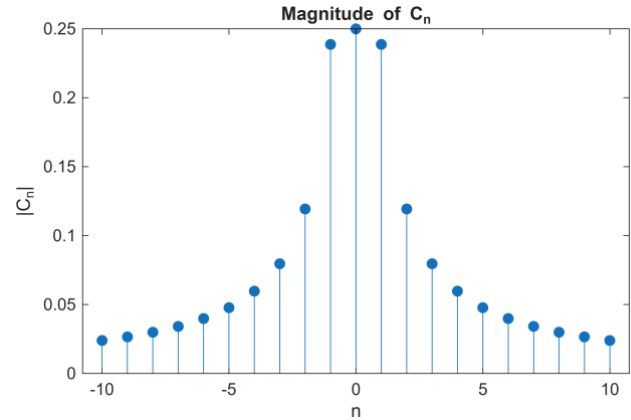


Figure 1. Magnitude spectrum of the exponential Fourier series coefficients for $-10 \leq n \leq 10$, illustrating peak energy concentration at the fundamental frequency.

3.2 Signal Reconstruction and the Gibbs Phenomenon

Reconstructing a signal that contains amplitude changes using a limited number of frequencies results in a behavior known as the Gibbs phenomenon. Generating the analytical expression for the truncated signal $x_{pN}(t)$ by utilizing the computed finite series of 21 harmonics produces algebraic complexity and the time-domain also needs numerical substitution over time vector $\Delta t = 0.001$. Overlaying this reconstructed sequence against the ideal original signal $x_p(t)$ shows the mathematical limitations of finite harmonic summation while the approximation accurately follows the linear downward slope of the sawtooth wave. This exhibits significant structural deviations at the interval boundaries.

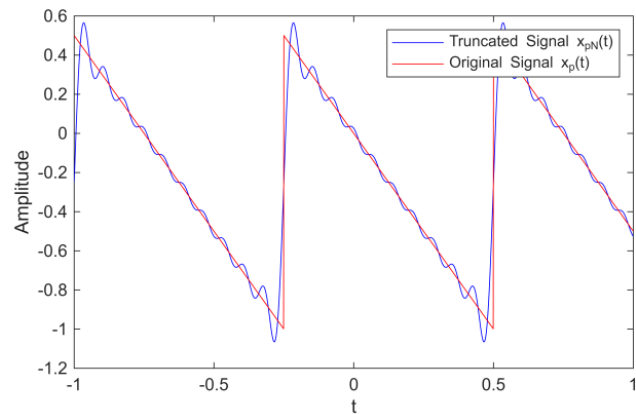


Figure 2. Time-domain comparison of the original sawtooth signal and its truncated Fourier reconstruction, highlighting oscillatory ringing at waveform discontinuities.

3.3 Transfer Function and Cut-off Linearity

A system's transfer function dictates its frequency filtering behavior and the 3dB cut-off frequency in the exercise defines the exact boundary where the transmitted signal loses half of its

power. For aperiodic continuous-time systems the exponential impulse response $h(t) = \frac{a}{2}e^{-a|t|}$ represents an ideal low-pass filter. Deriving the transfer function $H(\omega)$ using continuous Fourier transform establishes the baseline frequency-domain behavior. Analytically solving for the 3dB cut-off frequency yields multiple algebraic roots due to the quadratic nature of the frequency variable and getting the magnitude establishes the exact mathematical relationship $\omega_c = a\sqrt{\sqrt{2}-1}$. Evaluating and plotting that magnitude expression through $0 \leq a \leq 100$ generates a strictly positive linear plot. This graphical result mathematically proves a direct proportional dependency with a scaling constant of approximately 0.6436 where the operational frequency boundary of the filter is defined by its exponential scaling factor a .

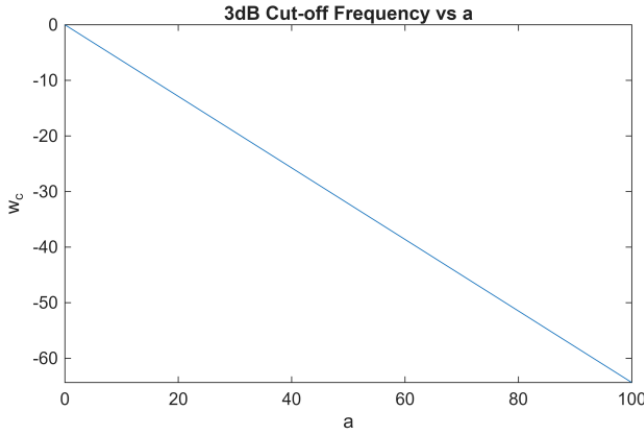


Figure 3. Linear proportional relationship between the exponential system parameter a and the resulting positive 3dB cut-off frequency magnitude ω_c .

3.4 Parametric Bandwidth Expansion

Increasing a low-pass filter's cut-off frequency naturally widens its passband that allowing a broader range of frequencies to pass through the system unattenuated. The practical implications of the previously established linear relationship are validated by evaluating the magnitude response curves $|H(\omega)|$ at discrete parametric intervals of $a = 10, 50, 80$, and 100 . As the defining parameter a scales upward the attenuation slopes relaxes causing the bell-shaped passband to widen uniformly outward across the frequency axis. Consequently, higher cut-off frequencies natively expand the system's absolute bandwidth permitting a broader continuous spectrum of input frequencies to propagate through the system unattenuated [3].

4 DISCUSSION AND CONCLUSION

The laboratory exercise successfully demonstrated the mathematical modeling, manipulation, and characterization of continuous-time signals and systems using MATLAB, fulfilling the primary objective to define and visualize these sequences. The first phase of the experiment confirmed that temporal manipulations, specifically time shifting, flipping, and scaling, directly alter a signal's independent variable to achieve predictable geometric changes. As evidenced in the results, the baseline signal $x(t)$, which originally spanned from $t = 0$ to $t = 8$ seconds, was successfully shifted, reflected, and compressed; for instance, the operation $x(2t)$ halved the signal duration to 4 seconds. These transformations validated the foundational theory of Linear

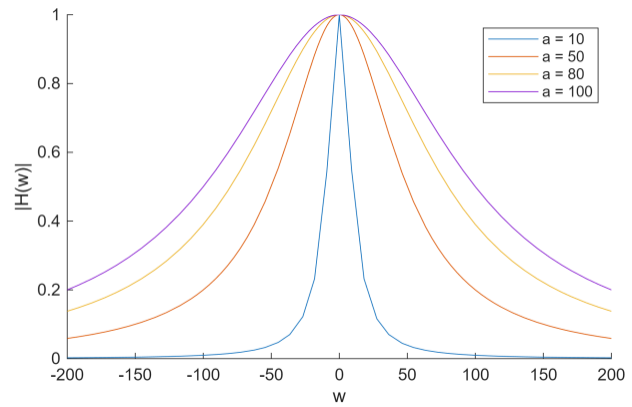


Figure 4. Superimposed magnitude responses demonstrating continuous bandwidth expansion as the exponential parameter a increases from 10 to 100.

Time-Invariant (LTI) systems by showing how signal geometry is modified through precise mathematical operations [1].

Furthermore, the characterization of systems through impulse responses $h(t)$ provided critical insight into filter behavior and frequency selectivity. By implementing symbolic equations for low-pass, highpass, bandpass, and bandstop filters, the experiment showcased how different systems react to an instantaneous unit impulse. A direct comparison between the lowpass and highpass responses revealed a notable symmetry: the lowpass response $h_1(t)$ exhibited a positive exponential decay starting at an amplitude of 1, while the highpass response $h_2(t)$, excluding the initial impulse, mirrored this with a negative decay starting at -1 . This practical application of the Dirac delta $\delta(t)$ and Heaviside step $u(t)$ functions in MATLAB reinforced the theoretical understanding of how LTI systems are entirely defined by their impulse responses [3].

Finally, the experiment explored internal signal analysis through the computation of auto-correlation $R_{xx}(\tau)$, which measured the signal's temporal similarity to itself. By transitioning from symbolic expressions to numerical data arrays with a high-resolution time step of $\Delta t = 0.01$, the total energy of the signal was identified at the zero-lag peak, reaching an amplitude of approximately 2.7. The resulting symmetric plot confirmed that as the time lag τ increases, the correlation drops, eventually returning to zero when the waveforms no longer overlap. In conclusion, the integration of MATLAB's computational tools effectively bridged theoretical concepts with practical simulation, providing a robust framework for analyzing the interaction between signals and systems [?].

5 AUTHORS CONTRIBUTION

On what contributions to specify, see the terms at www.elsevier.com.

- 1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
- 2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
- 3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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